

Technical Assessment 3: Case-Based Reasoning System Documentation

Course: CS0065 - Intelligent Systems

Case Study: Case Study 2: AI-Based Legal Decision Support System

A. Description of Case Base Representation

The past cases represent the basis against which new cases in the system are compared in reaching the desired conclusion. Each case has problem and solution factors.

Structure Definition

- **Problem Schema (problem):**
 - **charge** (*String*): Specific cybercrime classification/
 - **amount_stolen** (*Float/Int*): Financial value involved in Philippine Pesos (PHP).
 - **evidence_type** (*String*): Primary technical evidence provided by the prosecution (e.g., IP_address_logs, 2FA_bypass_logs, session_cookie_hijack).
 - **is_repeat_offender** (*Boolean*): Prior criminal history indicator.
- **Solution Schema (solution):**
 - **primary_defense** (*String*): Core defense argument pattern.
 - **bail_recommendation** (*Int*): Recommended bail amount in PHP.
 - **applicable_statute** (*String*): Relevant section of Philippine laws or the Cybercrime Prevention Act.

Initial Cases

Case ID	Charge	Amount Stolen	Evidence Type	Repeat Offender	Base Bail
LEGAL_2021_01	Unauthorized Bank Transfer	PHP 50,000	IP Address Logs	False	PHP 30,000
LEGAL_2022_05	Online Identity Theft	PHP 200,000	Phishing Site Logs	True	PHP 120,000
LEGAL_2023_12	Digital Fraud	PHP 300,000	Wire Transfer Records	True	PHP 180,000
LEGAL_2024_03	Unauthorized Bank Transfer	PHP 80,000	Session Cookie Hijack	False	PHP 45,000

B. Similarity Method (Retrieval Stage)

The retrieval stage identifies the past legal case that most closely matches a new client's situation using a **Weighted Nearest Neighbor Similarity Assessment** model. Because legal reasoning weighs legal charges more heavily than financial amounts, attributes are assigned proportional weights totaling 1.00

Feature Weights and Scoring Functions

1. **Charge Similarity (Weight: 0.50):** Categorical exact matching function.
1.0 if new_p["charge"] == past_p["charge"] else 0.0
2. **Amount Stolen Similarity (Weight = 0.30):** Normalized absolute difference function evaluated against the maximum scale (MaxRange = 300,000):
 $1 - (|\text{amount_new} - \text{amount_past}| - \text{MaxRange})$
3. **Offender History Similarity (Weight = 0.20):** Exact binary matching function for recidivism status.
1.0 if new_p["is_repeat_offender"] == past_p["is_repeat_offender"] else 0.0

C. Adaptation Rules

1. Bail is scaled proportionally based on the ratio between the new stolen amount and the retrieved precedent's stolen amount/
2. If the new client is a repeat offender (is_repeat_offender == True), a 1.5x penalty multiplier is applied to the calculated bail amount, and caution notes regarding bail challenges are appended to the defense strategy.
3. If the evidence type of the new problem differs from the retrieved case, the reasoning engine appends a forensic integrity clause targeting the specific digital evidence type directly into the primary_defense text string.

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NEW CLIENT CASE INPUT (PROBLEM)
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Charge       : unauthorized_bank_transfer
Amount Stolen : PHP 150,000
Evidence Type : 2FA_bypass_logs
Repeat Offender : False

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1. RETRIEVAL PHASE: EVALUATING PRECEDENT SIMILARITY
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-> Comparing with LEGAL_2021_01: Similarity Score = 90.00%
-> Comparing with LEGAL_2022_05: Similarity Score = 25.00%
-> Comparing with LEGAL_2023_12: Similarity Score = 15.00%
-> Comparing with LEGAL_2024_03: Similarity Score = 93.00%

RETRIEVED PRECEDENT : LEGAL_2024_03 (93.00% Match)

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2 & 3. REUSE & REVISE PHASES: ADAPTED LEGAL STRATEGY
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Primary Defense : Assert third-party malware infection without client awareness or intent. [ADAPTED: Specifically contest validity of '2FA_bypass_logs' via forensic integrity audit].
Recommended Bail : PHP 84,375
Applicable Statute : Cybercrime Prevention Act - Sec 4(a)(1) Illegal Access

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4. RETAIN PHASE: KNOWLEDGE BASE UPDATE
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Status       : New case successfully retained into Case Base[cite: 2].
Assigned Case ID : LEGAL_2026_05
Total Case Base : 5 cases stored.
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